College of Engineering

Department of Mechanical Engineering

ME 340 - Introduction to Manufacturing Engineering

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HAAS CNC Lathe Report

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Problem Definition

In accordance with shop safety guidelines, the team will produce a ring with external threads in accordance with the dimensions listed in the lab manual. A CNC lathe will be prepared and run over a 1” diameter aluminum bar in order to manufacture the part.

Deliverables

- A 0.5” long, 0.13” thick aluminum ring with 1-12 UNF external threads.
- A detailed report on the actions taken and results made using the CNC lathe.

Procedure

1. Place the aluminum rod in the chuck.
2. Load tools into their respective slots.
3. Home each tool by moving the dials in JOG mode until they are a paper thickness away from the part. Press OFFSET. Highlight the tool number. Press XYZ FACE MEASURE. In the same offset tab, type in the radius geometry after measuring with a caliper. Press ENTER.
4. Select TIP DIRECTION. Choose the "3 X- Z-" orientation. Press Enter
5. Repeat steps 3&4 for all tools with their respective dimensions.
6. To drill through, press MDI + edit to access VPS. Then press DRILL TAP followed by DRILLING. Press CYCLE START to run the operation.
7. Repeat step 6 for OD-turning, external threading, ID turning, and parting off while inputting the values from the tables in the lab manual.
8. When finished, open the doors, remove the finished part, spray and clean the area, debur the part, and return the tools to the rack.

Reflection

Operation of the lathe requires basic computer skills to traverse the digital interface. The only other required skill is attention to detail when setting up the offsets and loading the machine with tools and parts. Overall, the lab was very straightforward and easy to follow through the instructions.

Work



Figure 1: HAAS CNC Lathe with 4-station turret.



Figure 2: The main control panel on the lathe.



Figure 3: The 1” aluminum bar used as raw material for which the ring will be cut from.



Figure 4: The finished product with 1-12 UNF threads.

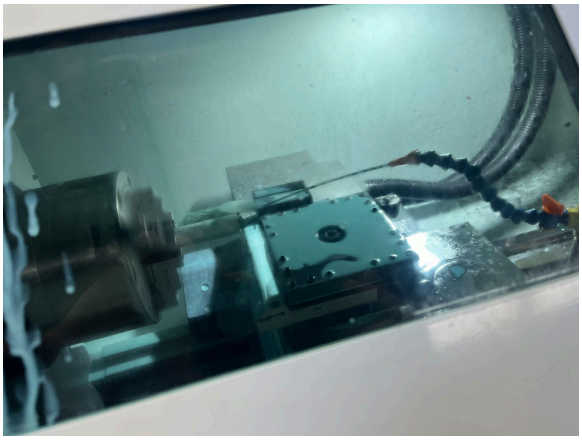


Figure 5: Visual of the operation in-progress.

Conclusion and Recommendations

1. Describe the results of the event programming and lathe labs, covering both software and hardware results.

   Initiating the offsets and traversing the computer interface was as simple and straightforward as following the directions and pressing a few buttons. The physical operation on the part required loading the raw material, waiting for the lathe to finish its program, and collecting the part. There were no major issues conducting either software or hardware procedures.

2. Discuss any errors or defects observed in the machined products during the initial operation and after improvements were made.

   Because the lathe carved out the inside of the ring before cutting it off from the rest of the aluminum rod, a lip was formed on the inside of the ring reducing the hollow diameter and creating a rough edge. However, this problem was easily solved using a file to shave off the extra material.

3. Describe any challenges encountered during the programming and operation, and explain how you resolved them.

   Two issues arose in the lab. Because some of the drill tips were angled inwards, the small distance had to be adjusted into the offsets which required additional measurements and re-typing in the home positions. The second issue occurred when the final cut formed a lip on the ring. This was worked through by employing a course file to shave off the edges and give it a smoother finish.

4. Identify potential areas for process improvement observed after the first operation session.

   The operating time could be cut down drastically if all the threads of the rod were cut at once followed by the hollowing process all at once, and then finally cut up into pieces. This would eliminate the need for the tools to rotate around and the user to pull out more of the aluminum rod after each individual part. The machine would also benefit from a catch basin so that the part does not accidentally fall between the mechanism to the bottom of the machine where it is hard to reach.

5. Analyze the actual process results, both quantitatively and qualitatively, and compare them with the intended outcomes. Discuss how closely the actual results matched the expected impacts and explain any discrepancies.

   The ring was exactly 0.5in wide with an outside radius of 0.48in as expected. However the inside diameter - which should have been 0.35in - was found to be 0.31 in after being pushed in due to the lip forming from the processing. The final finish on the edges of the part were jagged and sharp, but a sandpaper finish was able to fix the issue.